Cooling Environments

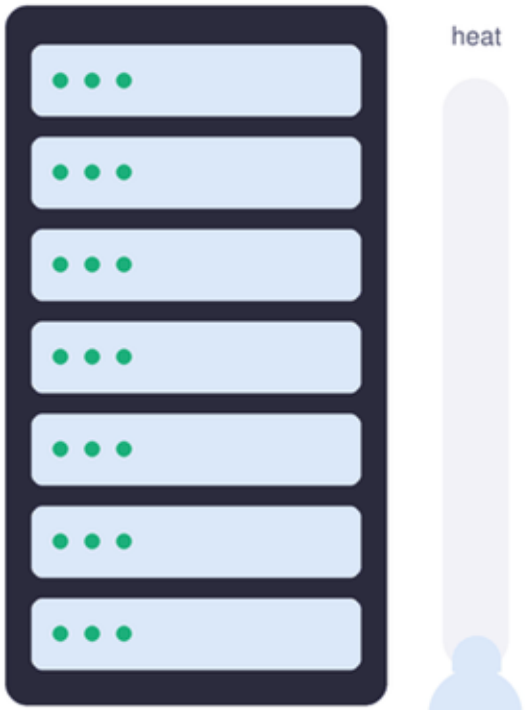
# Managing Liquid Cooling At-Scale: Tooling, Telemetry, and Reliability for High-Density AI Racks

Zach Miller *Hardware Systems Engineer, Meta*

Kalpak Dhake *Thermal Engineer, Meta*

OCP PLATINUM Member

# Power Density Growth



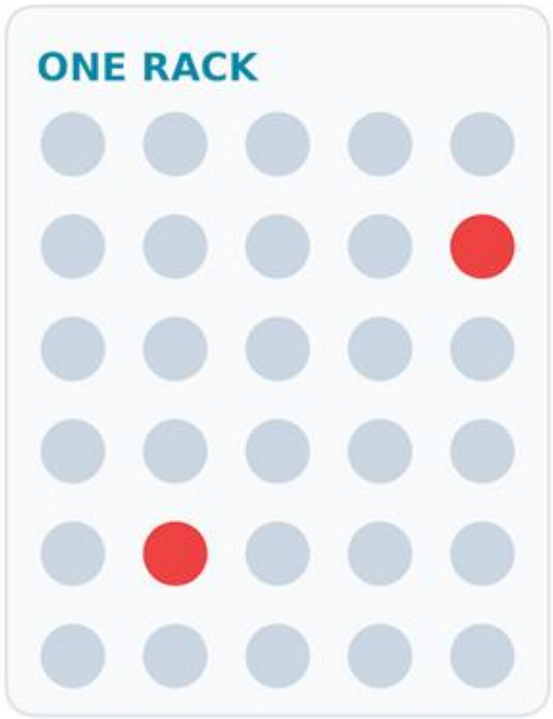

**12** kW / rack

**Air cooling: sufficient**

Cooling Type: Air

2015 2020 2026+

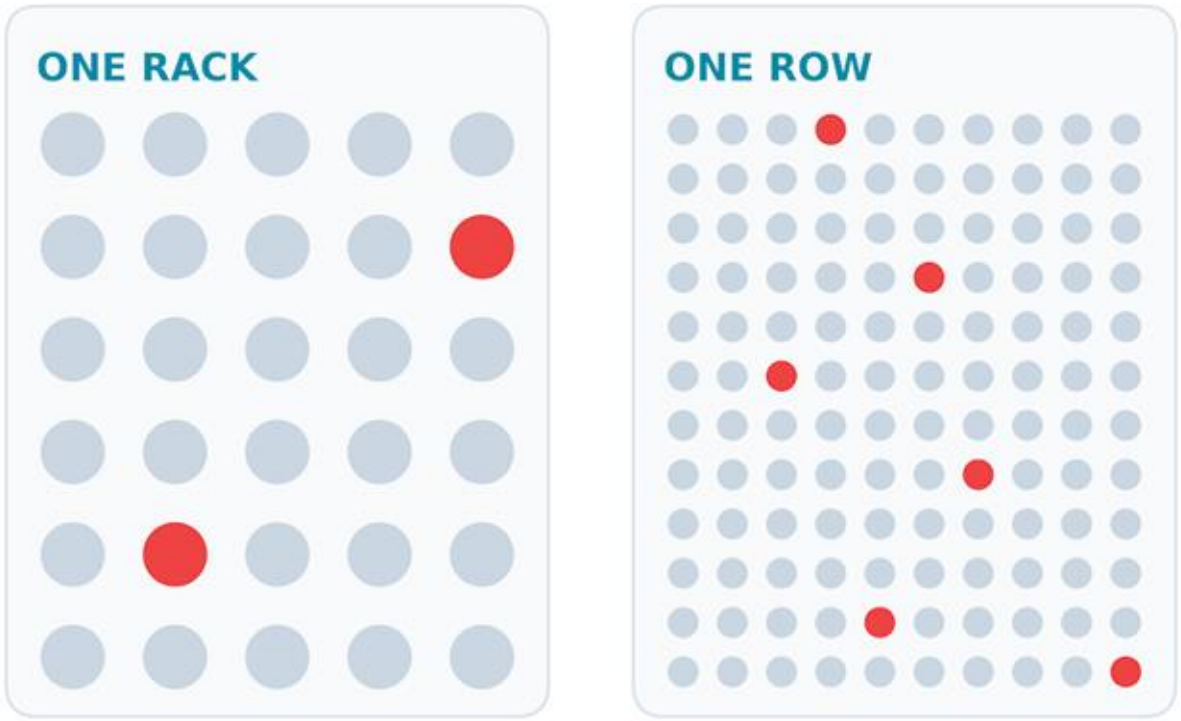
# Scaling Challenges




THE PROBLEM

**Failure points scale with the fleet;**
**Repair capacity does not**

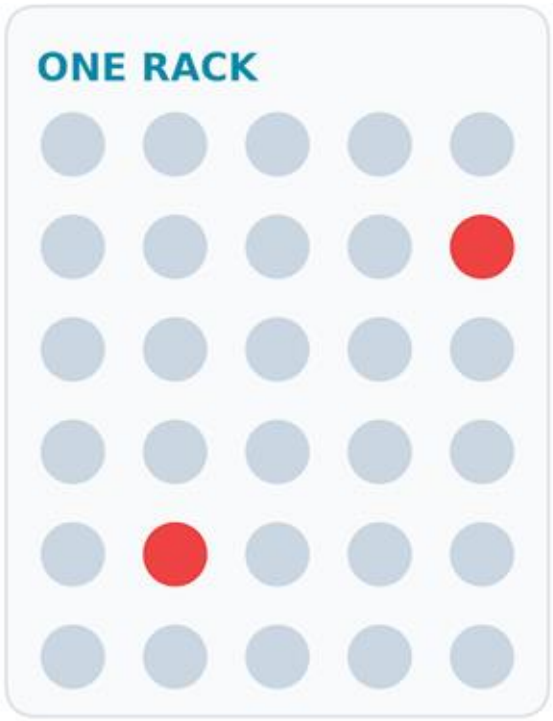
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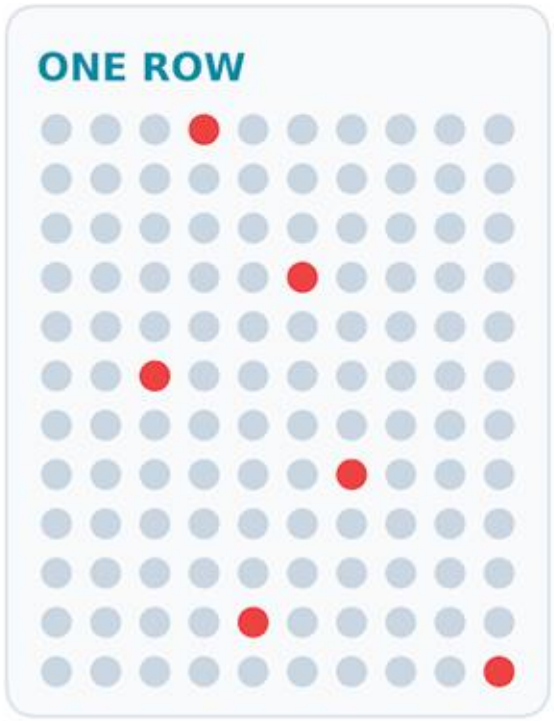


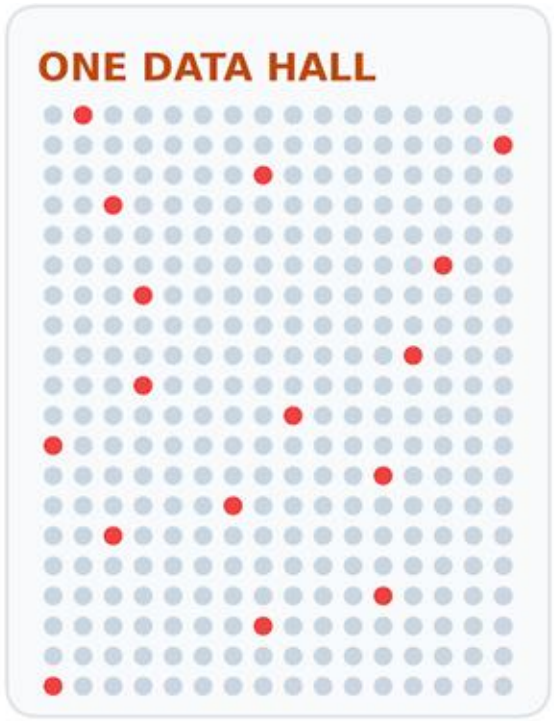

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# Scaling Challenges



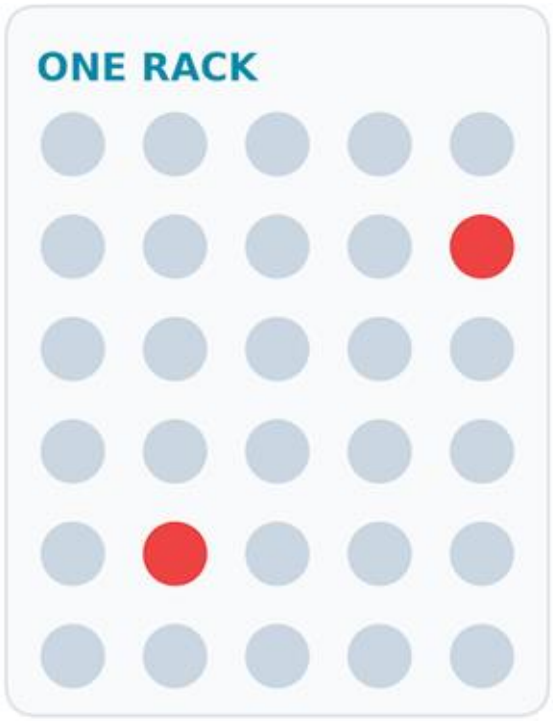



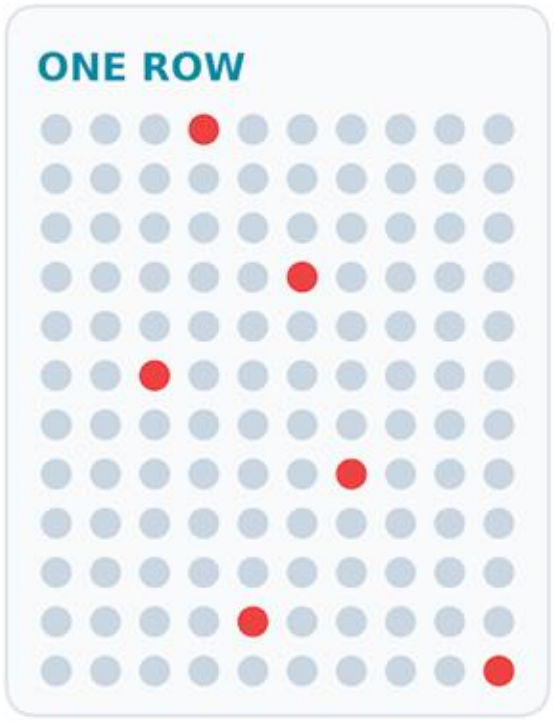



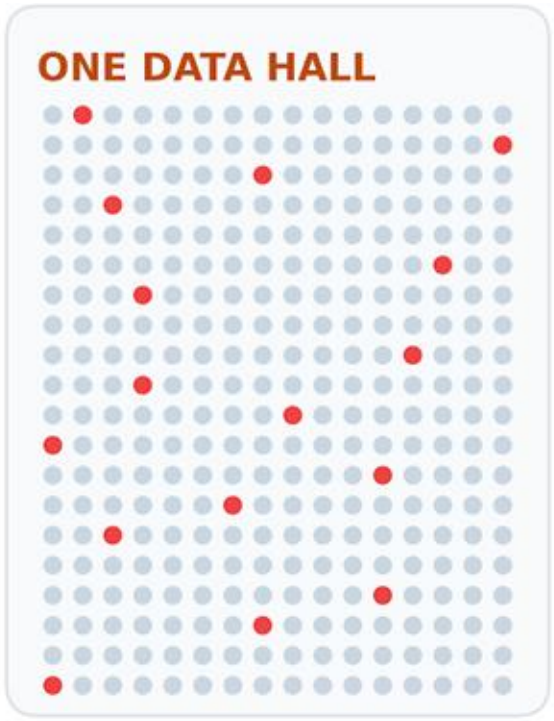

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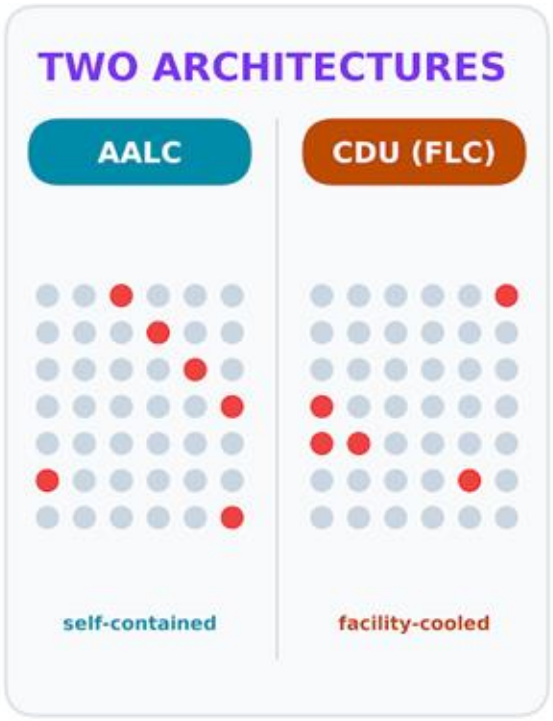
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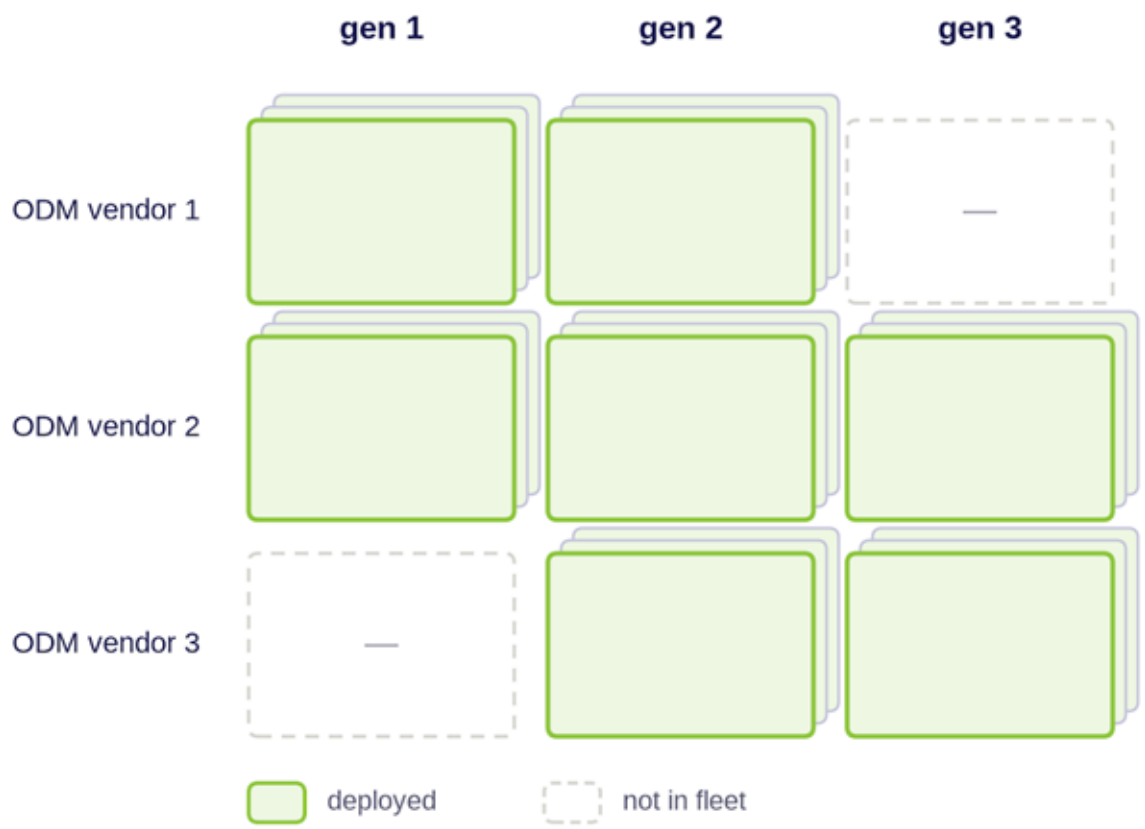

THE PROBLEM

**Failure points scale with the fleet;**
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# Monitoring at Scale

## Every filled cell is a different register map

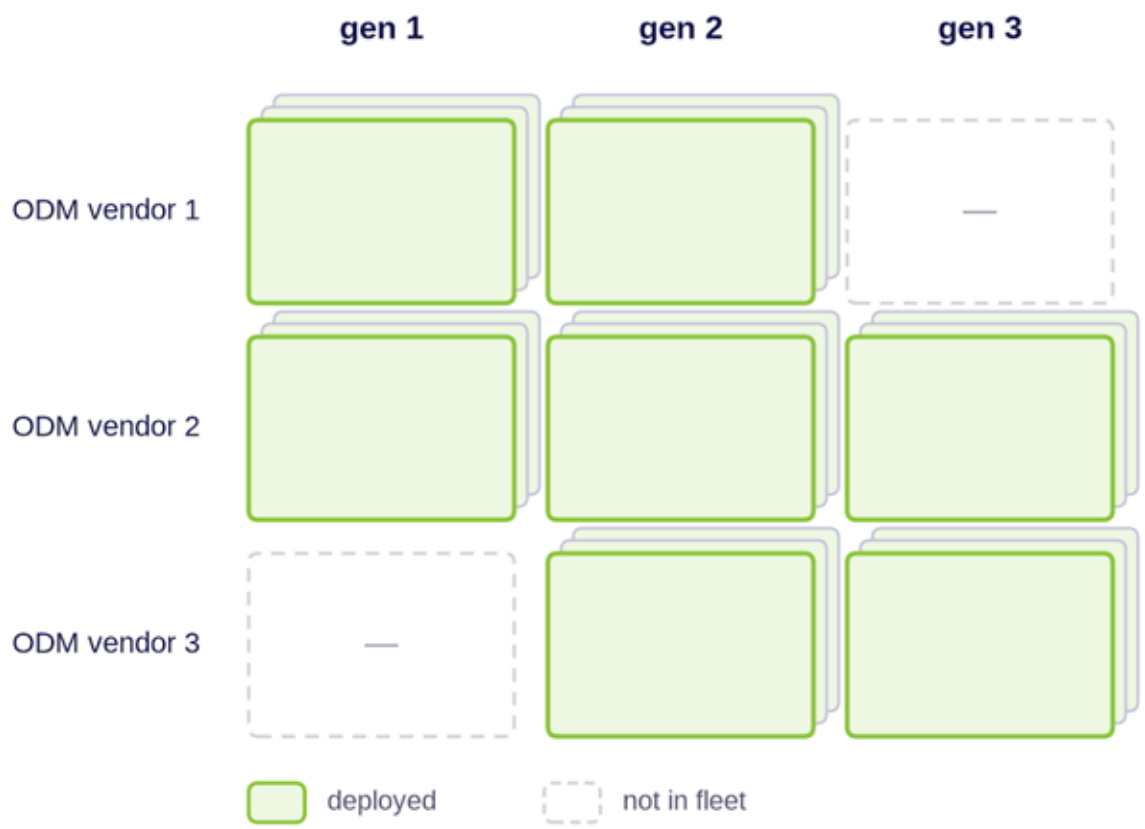
One fleet, many machines that are not the same



# Monitoring at Scale

## Every filled cell is a different register map

One fleet, many machines that are not the same




| | | |
|---|---|---|
| | A | hardware generations |
| x | B | ODM vendors |
| x | C | controller revisions |
| x | D | firmware versions in flight |

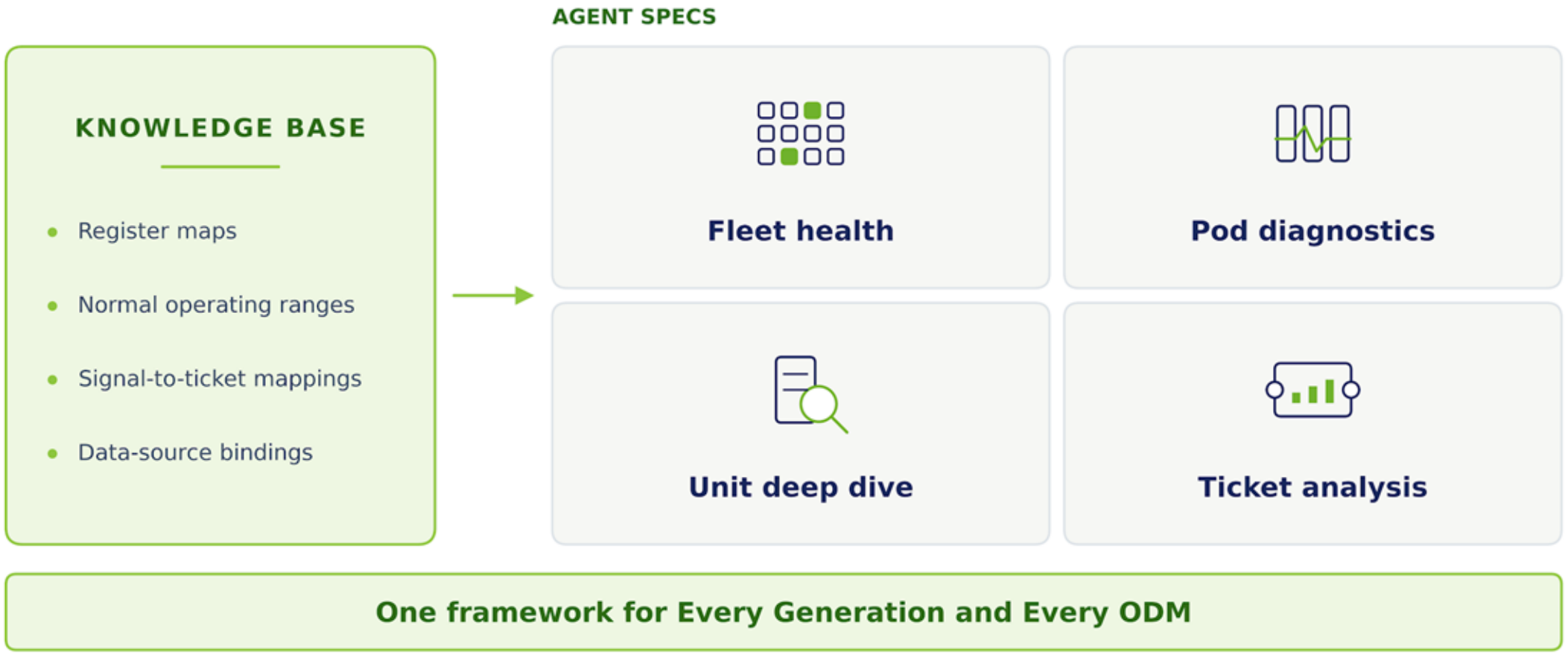
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**50+**

**distinct configurations in production**

# Monitoring at Scale

## Diagnosis as Code




**One framework for Every Generation and Every ODM**

# Monitoring Paradigms

**OUT-OF-BAND**

**RACK**
compute trays · liquid cooling · busbar PSUs · pumps · valves · sensors

**IN-BAND**

## 1 HARDWARE MANAGEMENT CONTROLLER

**Management Controller (BMC / RMC)**
Embedded controller physically on the rack, with its own MCU, firmware and standby power domain.

**Redfish**
DMTF-standard REST API exposed on the OOB interface. Chassis / Thermal / Power / Sensors resource tree.

**Sensor Telemetry**
Direct I2C / PMBus / Modbus polling of the hardware: temperature, flow, pressure, leak GPIO, fan and pump RPM.

**Power Handling & Actuation**
Power on / off / cycle, throttle and emergency shutdown. Direct actuation of PSUs, pumps and valves.

## 2 DATACENTER TOOLING

**Host Agents**
In-band software running on the host OS inside the rack, deployed and versioned with the fleet image.

**Data Collection Daemons**
Scrape counters, logs, driver and kernel state; publish to time-series and event pipelines on a fixed cadence.

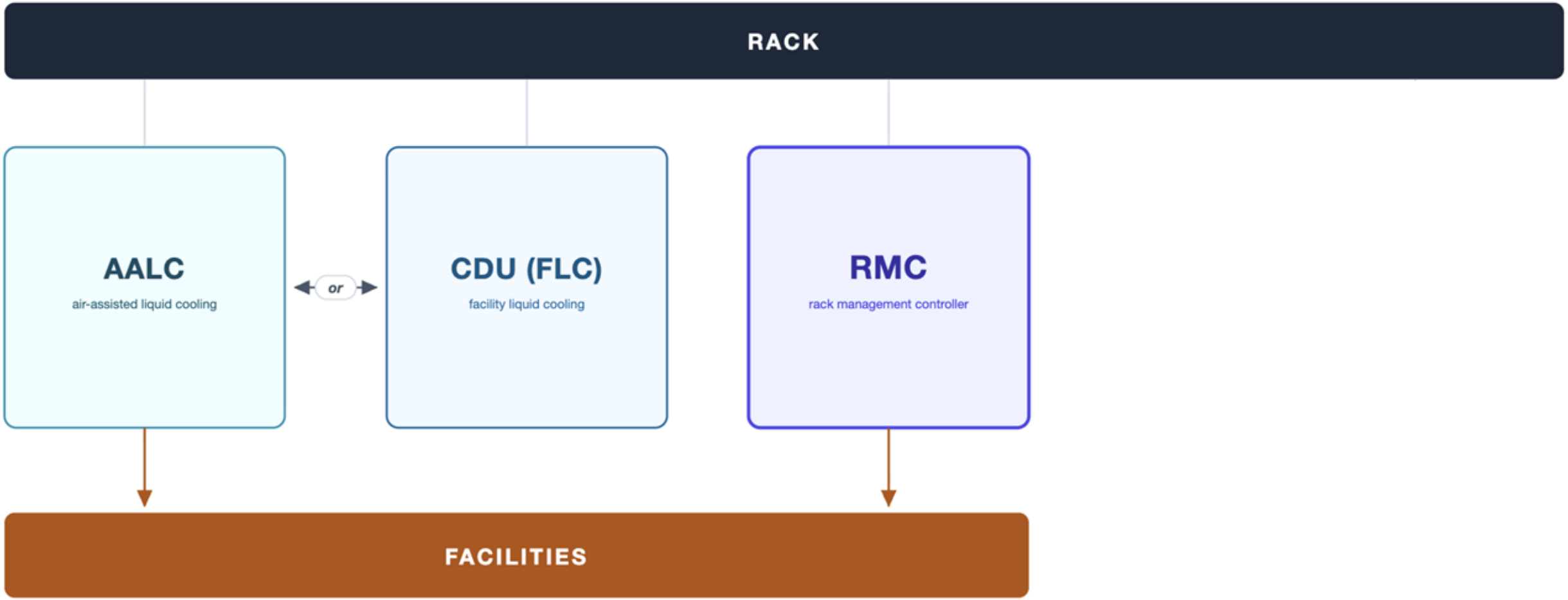
**DC Network & API Connections**
Routable datacenter network path to and from the rack; Thrift / gRPC / REST service calls between tiers.

**Fleet Management Platform**
Orchestrates health checks across the fleet and drives automated remediation workflows and repair tickets.

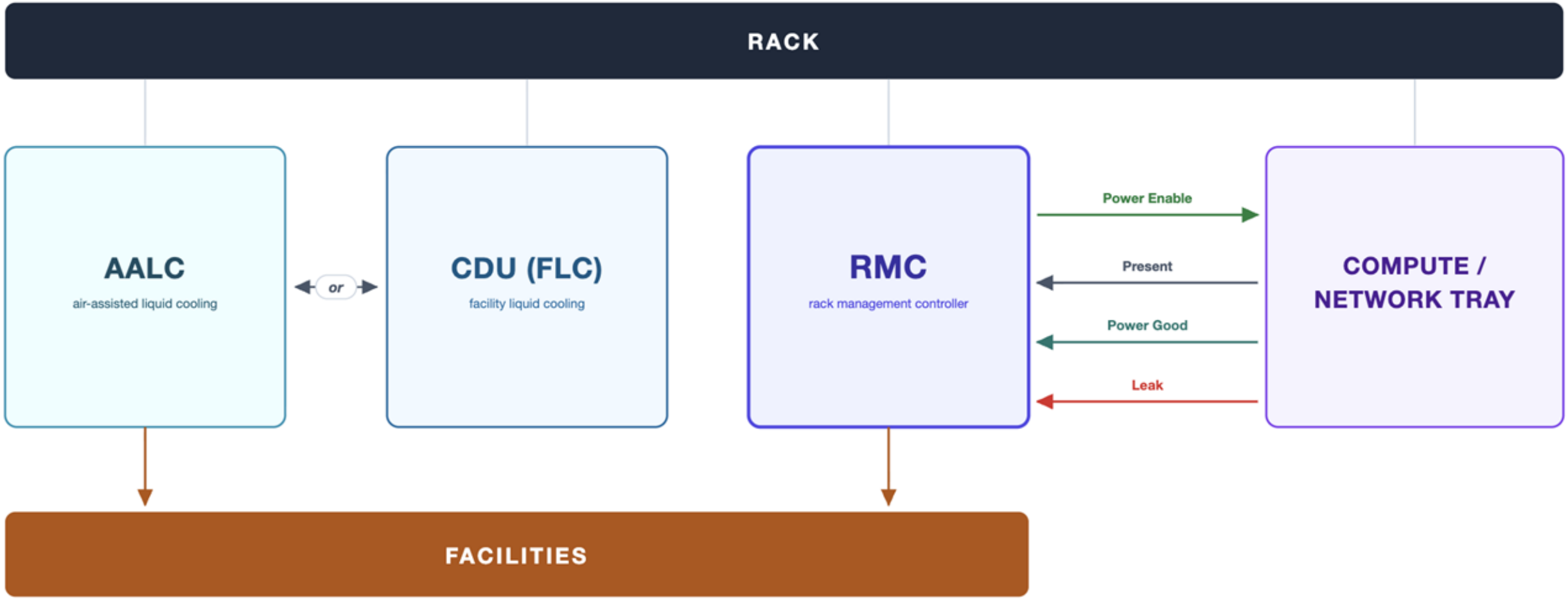
# Hardware-Based Monitoring



# Hardware-Based Monitoring



# Hardware-Based Monitoring



# Automation Monitoring

**LIQUID COOLING**
sensors

- Temperature
- Flow rate
- Duty cycle
- Pressure
- Leak state

**IT RACK**
IT metrics

- GPU temperature
- Power draw
- Cold plate ΔT
- Throttle events

# Automation Monitoring

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IT metrics

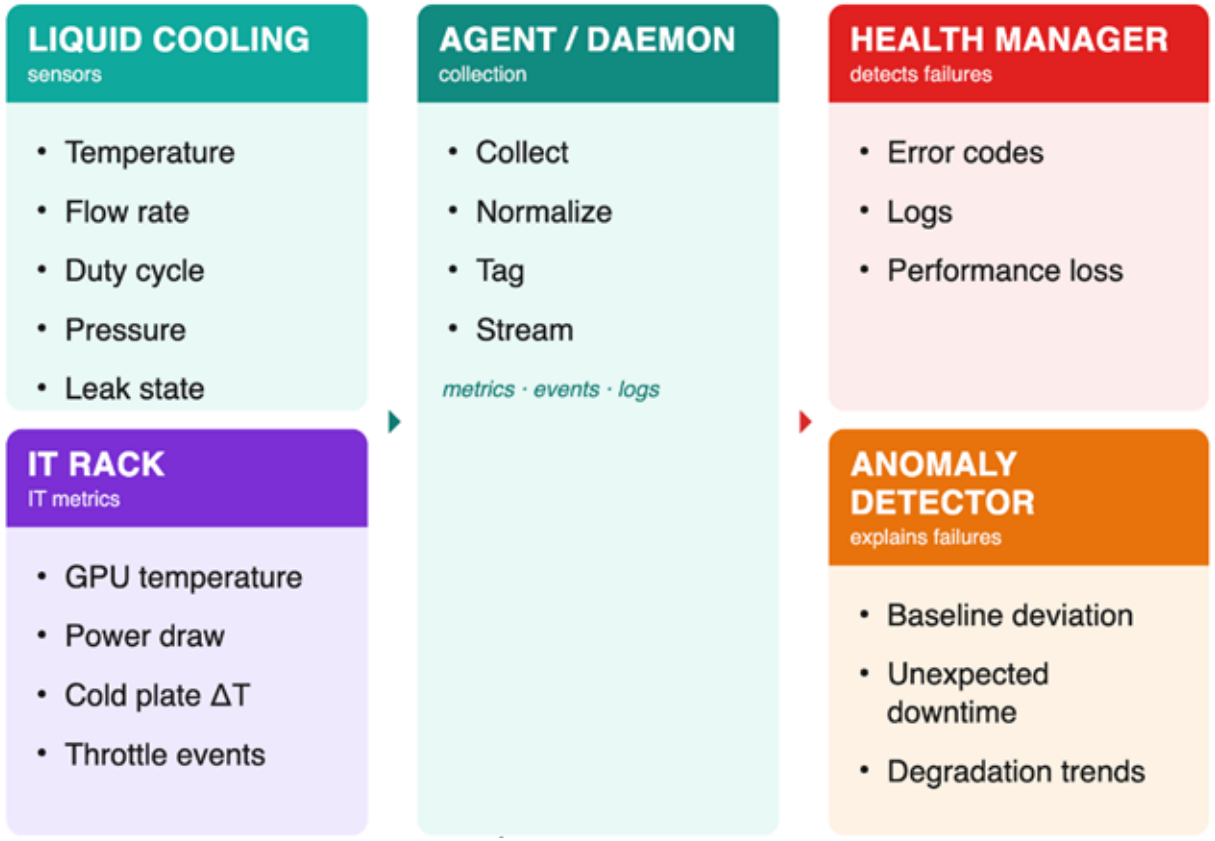
- GPU temperature
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**AGENT / DAEMON**
collection

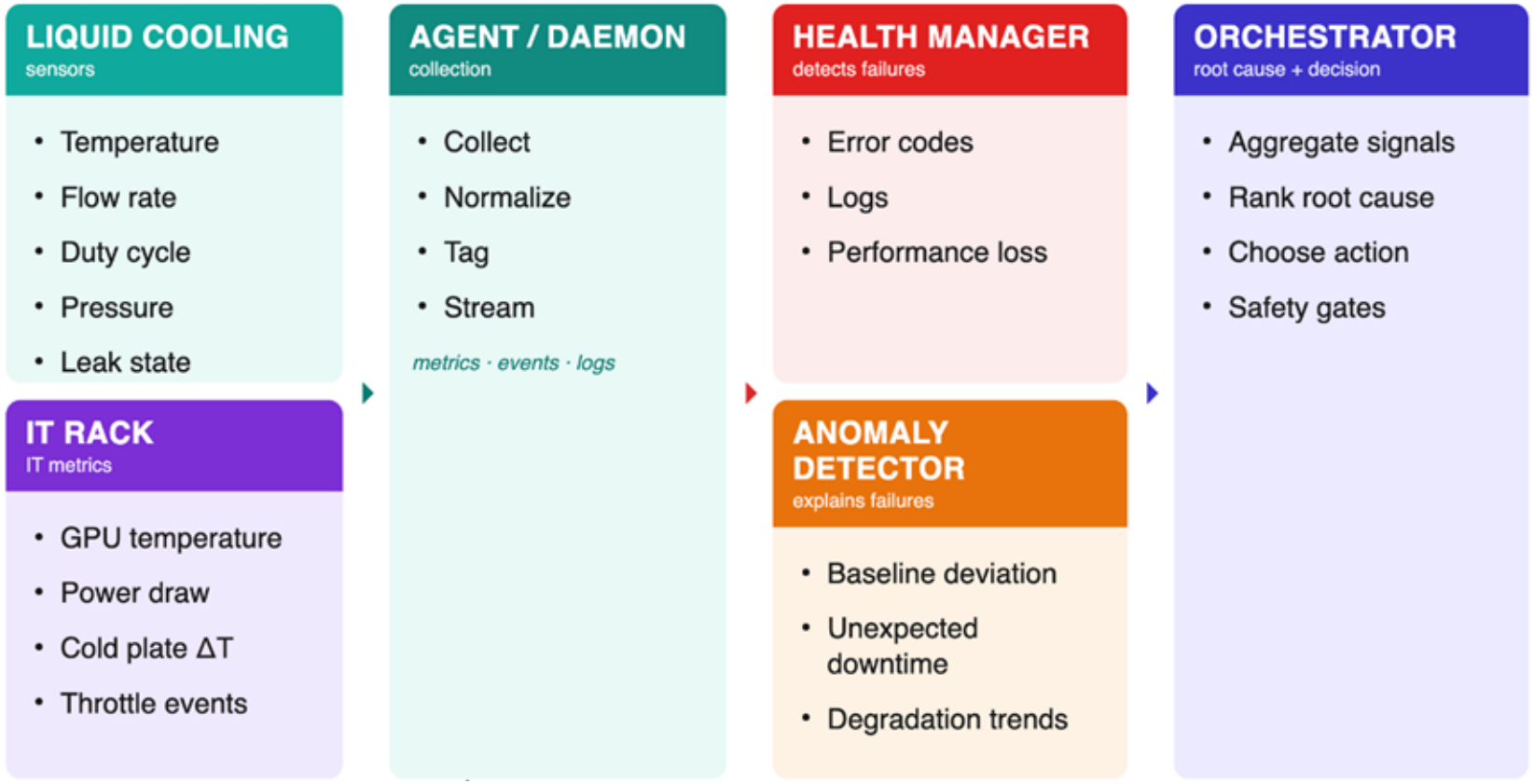
- Collect
- Normalize
- Tag
- Stream

*metrics · events · logs*

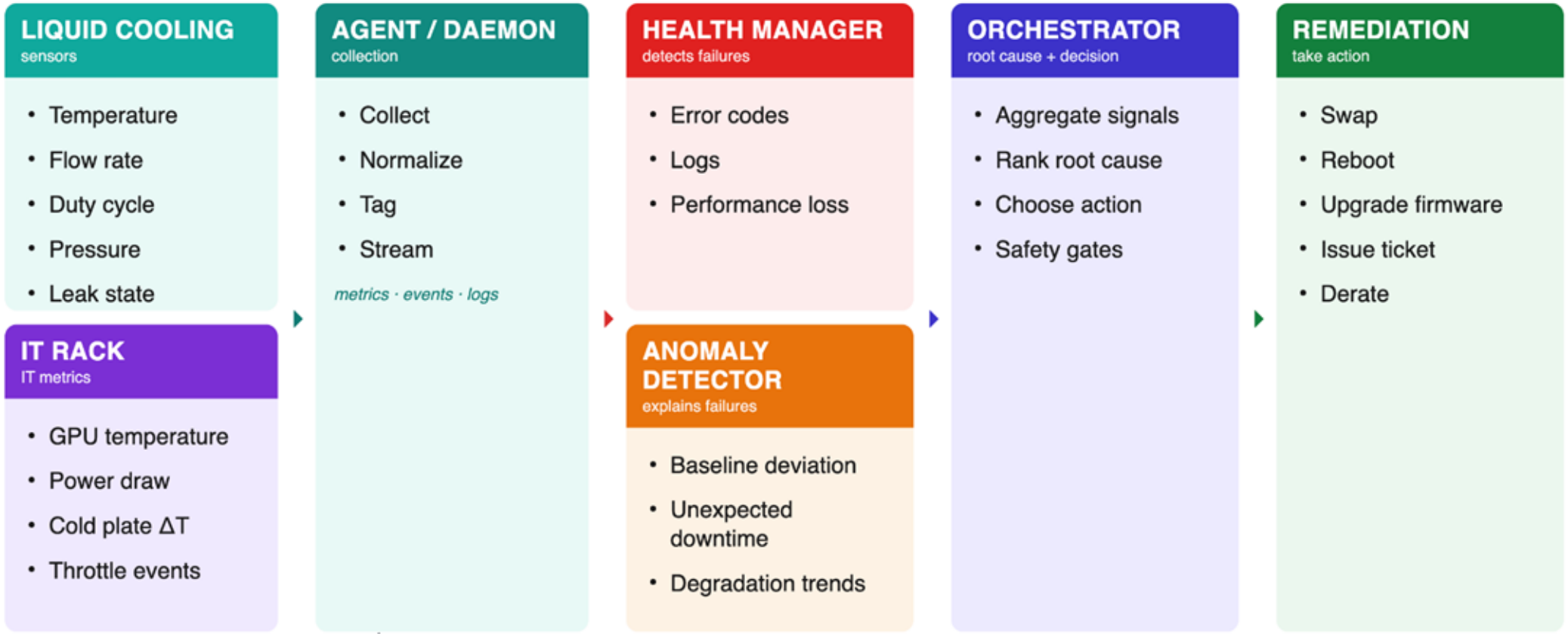
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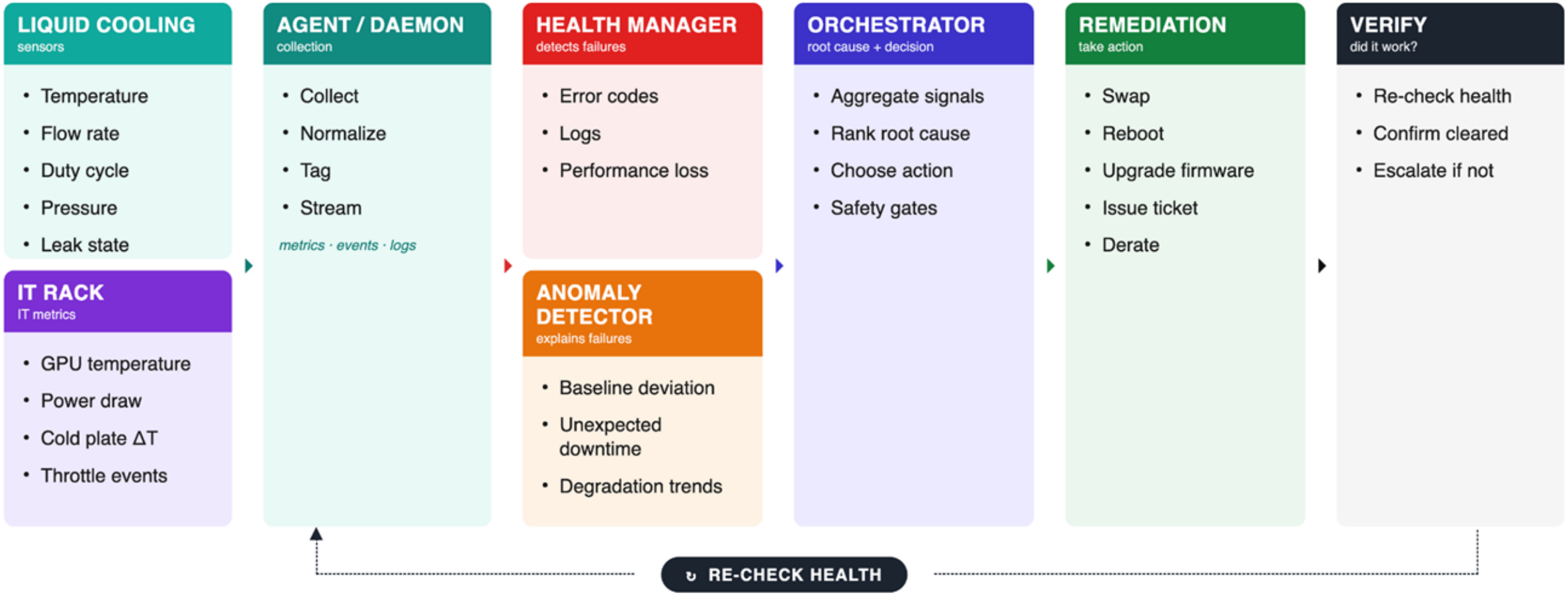
# Automation Monitoring



# Automation Monitoring



# Automation Monitoring



# Facility Data Monitoring

**IT / RACK DOMAIN**

**LIQUID COOLING**

Temperature
Flow rate
Duty cycle
Leak state

**IT HARDWARE**

GPU temperature
Power draw
Cold plate ΔT
Throttle events

**IN-RACK AGENT**

fast sampling
per-rack scope

# Facility Data Monitoring

## IT / RACK DOMAIN

### LIQUID COOLING
- Temperature
- Flow rate
- Duty cycle
- Leak state

### IT HARDWARE
- GPU temperature
- Power draw
- Cold plate ΔT
- Throttle events

### IN-RACK AGENT
fast sampling
per-rack scope

## FACILITY DOMAIN

### ENVIRONMENTAL
- Temperature
- Humidity
- Dew point
- Air pressure

### FACILITY WATER
- Supply / return temp
- Loop flow
- CDU approach temp
- Plant status

### FACILITY COLLECTOR
slow sampling
hall-wide scope

# Facility Data Monitoring

## IT / RACK DOMAIN

### LIQUID COOLING
- Temperature
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- GPU temperature
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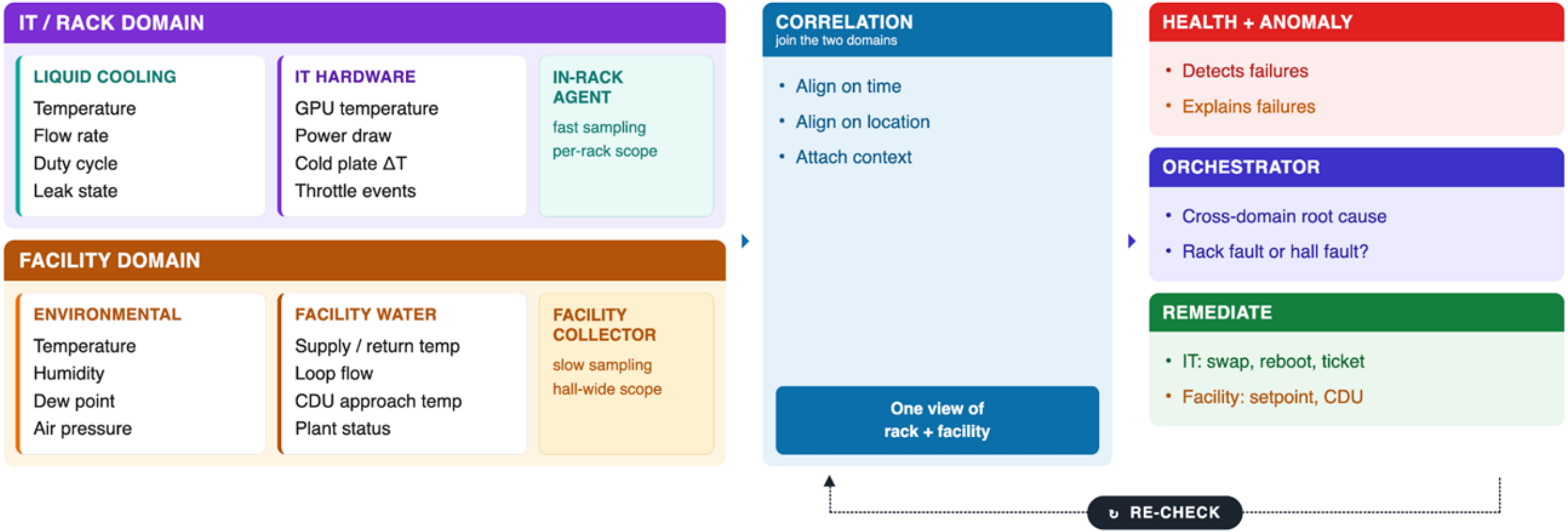
### FACILITY COLLECTOR
slow sampling
hall-wide scope

## CORRELATION
join the two domains

- Align on time
- Align on location
- Attach context

**One view of rack + facility**

# Facility Data Monitoring



# Three Incidents, One Pattern

| | **Humidity false alarms** |
|---|---|
| **SIGNAL WE SAW** | **Numerous racks** asserted leak and powered down. |
| **WHAT IT ACTUALLY WAS** | **Condensation** on the rope — rack surfaces fell below dew point. |
| **NOW AUTOMATED** | Debounce + cross-sensor filter: **~100x fewer events.** |

# Three Incidents, One Pattern

| | Humidity false alarms | Firmware misconfig |
|---|---|---|
| **SIGNAL WE SAW** | **Numerous racks** asserted leak and powered down. | **Thermal excursions** in stress test. Telemetry healthy, no fault. |
| **WHAT IT ACTUALLY WAS** | **Condensation** on the rope — rack surfaces fell below dew point. | **Generic default** pump duty, below platform requirement. |
| **NOW AUTOMATED** | Debounce + cross-sensor filter: **~100x fewer events.** | Duty in **version-controlled config**; ID written on every swap. |

# Three Incidents, One Pattern

| | Humidity false alarms | Firmware misconfig | Manifold leak |
|---|---|---|---|
| **SIGNAL WE SAW** | **Numerous racks** asserted leak and powered down. | **Thermal excursions** in stress test. Telemetry healthy, no fault. | **Coolant on the floor**, found on walkthrough. No alarm fired. |
| **WHAT IT ACTUALLY WAS** | **Condensation** on the rope — rack surfaces fell below dew point. | **Generic default** pump duty, below platform requirement. | **A real leak** at the rack rear, tracking down the frame. |
| **NOW AUTOMATED** | Debounce + cross-sensor filter: **~100x fewer events.** | Duty in **version-controlled config**; ID written on every swap. | **Shutdown authority moved into tooling.** |

# Call to Action

- Every liquid-cooling unit exposes the same physical quantities through a different register map, so operators re-implement detection, diagnosis and repair per vendor and per generation
- How to get involved
  - OCP Cooling Environment section
  - Contribute to the Machine-readable diagnostic spec (our agent definitions can be reference implementation)
  - A minimum telemetry contract for liquid-cooling units: same signals, same addressing, every vendor

# Thank You!

